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Global boosting circuit

Abstract

Systems and methods are provided for a memory device including a first memory array, a local input/output (LIO) circuit, and a global input/output (GIO) circuit. The first memory array includes a memory cell and a local bit line. The LIO circuit is configured to receive a local bit line signal on the local bit line and to generate a global bit line signal on a global bit line based on the local bit line signal. The GIO circuit is coupled to the LIO circuit and is configured to receive the global bit line signal. The GIO circuit comprises a latch circuit including a global bit line signal latch that is configured to latch the global bit line signal, and a booster circuit that is configured to drive the global bit line signal in the GIO circuit based on a previous global bit line signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/396,650, filed Aug. 10, 2022, entitled "Semiconductor Device and Method of Operating the Same," and U.S. Provisional Application No. 63/384,621, filed Nov. 22, 2022, entitled "Global Boosting Circuit for Read Global Bitline for Multi Bank Memories," both of which are incorporated herein by reference in their entireties.

FIELD
(1) The present disclosure relates to memory devices and in particular to multibank memories.
BACKGROUND
(2) Memory devices are electronic data storage devices that include memory banks that include memory locations for storage of data. Devices that include multiple memory banks may experience timing issues related to signals within those devices. Example memory banks may comprise a local input/output (LIO) circuit and one or more memory arrays. The memory bank may be coupled to a global input/output (GIO) circuit that generates a global input/output signal. For example, some memory devices include a plurality of memory banks. In such memory devices, the distance between the LIO circuits and the GIO circuit increases. This can lead to a greater time delay from generating signals (e.g., a global bit line signal) in the LIO circuit and receiving them in the GIO circuit. Furthermore, the greater time delay can require charging the signals in the LIO circuit for a significant amount of time to ensure the signals are latched in the GIO circuit. Therefore, there is a need in the art to mitigate or eliminate the dependency of latching the signals in the GIO circuit on charging the signals in the LIO circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following detailed description will be better understood when read in conjunction with the appended drawings. For the purpose of illustration, there is shown in the drawings certain embodiments of the present disclosure. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate an implementation of systems and apparatuses consistent with the present invention and, together with the description, serve to explain advantages and principles consistent with the invention.
(2) FIG. **1** depicts a block diagram of an example memory device, in accordance with some embodiments.
(3) FIG. **2** depicts further aspects of the example memory device, in accordance with some embodiments.
(4) FIG. **3** depicts a latch circuit and booster circuit, in accordance with some embodiments.
(5) FIG. **4** depicts a timing diagram of a read operation of a global bit line, in accordance with some embodiments.
(6) FIG. **5** depicts a method, in accordance with some embodiments.
DETAILED DESCRIPTION
(7) The following detailed description is provided to assist the reader in gaining a comprehensive

understanding of the methods, apparatuses, and/or systems described herein. Accordingly, various changes, modifications, and equivalents of the systems, apparatuses and/or methods described herein will be suggested to those of ordinary skill in the art. Also, descriptions of well-known functions and constructions may be omitted for increased clarity and conciseness.

(8) It is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. For example, the use of a singular term, such as, “a” is not intended as limiting of the number of items. Also the use of relational terms, such as but not limited to, “top,” “bottom,” “left,” “right,” “upper,” “lower,” “down,” “up,” “side,” are used in the description for clarity and are not intended to limit the scope of the invention or the appended claims. Further, it should be understood that any one of the features can be used separately or in combination with other features. Other systems, methods, features, and advantages of the invention will be or become apparent to one with skill in the art upon examination of the detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, be within the scope of the present invention, and be protected by the accompanying claims.

(9) Example memory devices include one or more memory banks comprising a local input/output (LIO) circuit and one or more memory arrays. A memory bank may be coupled to a global input/output (GIO) circuit that generates a global input/output signal. Some memory devices include a plurality of memory banks. In such memory devices, the distance between the LIO circuits and the GIO circuit increases. This can lead to a greater time delay from generating signals (e.g., a global bit line signal) in the LIO circuit and receiving them in the GIO circuit. Furthermore, the greater time delay can require charging the signals in the LIO circuit for a significant amount of time to ensure the signals are latched in the GIO circuit. This can delay the charging of the signals for a later clock cycle and can increase the time to access data from the memory device. Therefore, a memory circuit may benefit from mitigation or elimination of any dependency of latching the signals in the GIO circuit on charging the signals in the LIO circuit.

(10) In embodiments, it may be desirable to use components and circuits within the GIO circuit to drive the signals generated in the LIO circuit to assist in latching the signals in the GIO circuit. Embodiments disclosed herein involve using circuits in the GIO circuit to drive the signals generated in the LIO circuit. Examples and embodiments disclosed herein can reduce the access time of generated data and reduce or eliminate the dependency of latching the signals generated in the LIO circuit on charging the signals by components within the LIO circuit.

(11) FIG. **1** depicts a block diagram of an example memory device **100**, in accordance with some embodiments. The memory device **100** includes a memory cell array **101**, a local input/output (LIO) circuit **102**, and a global input/output (GIO) circuit **103**. The memory cell array **101**, which is an SRAM memory array in some examples, generates data signals **106** that are received by the LIO circuit **102**. The memory cells may include a plurality of transistors (e.g., a **4**T, **6**T, **8**T, or **10**T memory cell). The LIO circuit **102** receives the data signals **106** and generates a global bit line signal **107** on a global bit line. The GIO circuit **103** receives the global bit line signal **107**. The GIO circuit **103** includes a latch circuit **104** that latches the global bit line signal **107** in the GIO circuit **103**. The GIO circuit **103** further includes a booster circuit **105** configured to drive the global bit line signal **107** in the GIO circuit **103** based on a previous global bit line signal **107** from a previous clock cycle. In embodiments of the present disclosure, the booster circuit **105** and the latch circuit **104** within the GIO circuit **103** can reduce the access time of data generated within the memory device **100**, as compared with conventional memory devices. Furthermore, the booster circuit **105** and latch circuit **104** can reduce or eliminate the dependency of latching the signals generated in the LIO circuit **102** on charging the signals by components within the LIO circuit **102**.

(12) FIG. **2** depicts further aspects of the example memory device, in accordance with some embodiments. In the example embodiment depicted in FIG. **2**, the memory cell array **101** of the memory device **100** includes a plurality of memory cells **218**. The plurality of memory cells **218**

forms a column of memory cells. The column of memory cells are connected to the LIO circuit **102** by a bit line **203** and a complimentary bit line **204**. The bit line **203** receives a bit line signal from the plurality of memory cells **218**. The complimentary bit line **204** receives a complimentary bit line signal from the plurality of memory cells **218**. The complimentary bit line **204** and the bit line **203** include one or more column multiplexers **226** and read multiplexers **227**, which can be used to read bit data from a plurality of the memory cells **218** onto the bit line **203** and the complimentary bit line **204**. The LIO circuit **102** includes a sense amplifier **205**. The bit line **203** and complimentary bit line **204** are coupled to the sense amplifier **205**. The sense amplifier **205** receives the bit line signal, the complimentary bit line signal, a sense amplifier enable signal SAE **209**, and a sense amplifier pre-charge signal SAPREB **208**.

(13) The sense amplifier **205** is configured to generate a read bit line signal RBL **207** on a read bit line terminal and a read bit line bar signal RBLB **206** on a read bit line bar terminal based on the received bit line signal, complimentary bit line signal, and sense amplifier enable signal SAE **209**. For example, prior to a read operation, the sense amplifier pre-charge signal SAPREB **208** is enabled and charges the read bit line bar signal RBLB **206** and read bit line signal RBL **207** to a logic high (e.g., “1”). The sense amplifier **205** compares the voltage levels of the bit line **203** and the complimentary bit line **204** and reads bit data from a memory cell within the plurality of memory cells **218** based on the difference in voltage levels of the bit line **203** and the complimentary bit line **204**. For example, if the bit data in the memory cell is a logic high, the sense amplifier **205** may read a logic high based on the voltages of the bit line **203** and complimentary bit line **204** to which the memory cell is coupled.

(14) During a read operation, the sense amplifier enable signal SAE **209** is enabled to logic high. Based on the sense amplifier enable signal SAE **209**, the sense amplifier **205** amplifies the bit data from the bit line **203** and the complimentary bit line **204** to sufficient readable logic levels (e.g., logic high or “1”). The sense amplifier **205** generates the read bit line signal RBL **207** and the read bit line bar signal RBLB **206** based on the amplified bit data. For example, if the sense amplifier **205** reads a logic high signal based on the difference in voltage levels of the bit line **203** and the complimentary bit line **204**, the sense amplifier **205** may output a read bit line signal RBL **207** that is at logic high and a read bit line bar signal RBLB **206** that is at logic low.

(15) The read bit line signal RBL **207** is received by the inverter **210**, which outputs a read bit line NOT (RBLN) signal **219** that is the inverse of the read bit line signal RBL **207**. The RBLN signal **219** is received by the gate terminal of an NMOS transistor **212**. The NMOS transistor **212** is a pull-down transistor, which selectively pulls the read global bit line signal **107** to a VSS voltage level **220** (e.g., logic low) based on receiving a logic high RBLN signal **219** at its terminal. For example, if the read bit line signal RBL **207** is logic low, then the inverter **210** will output a logic high RBLN signal **219** and the NMOS transistor **212** will pull the read global bit line signal RGBL **107** to a VSS voltage level **220**. The read bit line bar signal RBLB **206** is received at a PMOS transistor **211** that is coupled to the NMOS transistor **212**. The read global bit line signal RGBL **107** is generated on a global bit line **213** that is coupled to a junction of the PMOS transistor **211** and the NMOS transistor **212**. The PMOS transistor **211** will selectively pull the global bit line signal RGBL **107** to a VDD voltage level **221** (e.g., logic high) based on receiving a logic low read bit line bar RBLB signal **206**.

(16) As shown in FIG. **2**, the global bit line **213** may pass through another memory array **214** including a second plurality of memory cells (not shown). The memory arrays and the LIO circuit **102** form a memory bank **222**. In a multi-bank memory device, the memory device **100** includes a plurality of memory banks **222**. For example, a plurality of memory banks **222** may be coupled to one another in a horizontal direction (e.g., a direction perpendicular to the global bit line **213**) and may form a memory row. Furthermore, a multi-bank memory may include a plurality of memory rows coupled to one another in a direction parallel to the global bit line **213**.

(17) The GIO circuit **103** is coupled to the memory bank **222** and the global bit line **213**. The GIO

circuit **103** receives the global bit line signal RGBL **107** on the global bit line **213**. In the example embodiment depicted in FIG. **2**, the GIO circuit **103** includes a multiplexer **216**. The multiplexer **216** receives the global bit line signal RGBL **107** and one or more redundant global bit line signals **223**. The redundant global bit line signals **223** include the same data as the global bit line **213** and are used to maintain the global bit line signal RGBL **107** if it becomes decoupled from the multiplexer **216**. For example, if the multiplexer **216** fails to read the global bit line signal RGBL **107**, the multiplexer **216** selects from a redundant global bit line signal **223** to maintain operation of the GIO circuit **103** and memory device **100**. The multiplexer **216** generates a global IO bar signal QB **224** based on the received global bit line signal RGBL **107** and redundant global bit line signals **223**. The global IO bar signal QB **224** is received at an inverter **217**, which outputs a global IO signal Q **225**. The GIO circuit **103** further includes a latch circuit **104** and a booster circuit **105** coupled to the global bit line **213**. The latch circuit **104** and the booster circuit **105** are discussed further with reference to FIG. **3**.

(18) FIG. **3** depicts a latch circuit **104** and booster circuit **105**, in accordance with some embodiments. As shown in FIG. **3**, the latch circuit **104** and booster circuit **105** are coupled to the global bit line **213**. The latch circuit **104** includes a global bit line signal latch **301**, a global bit line bar signal latch **302**, and a secondary latch **303**. A booster circuit **105** is coupled to the latch circuit **104**. The global bit line signal latch **301** is configured to latch (e.g., store) the global bit line signal RGBL **107**. The global bit line bar signal latch **302** is configured to latch a global bit line bar signal RGBLB **320** (e.g., an inverse of the global bit line signal RGBL **107**). The secondary latch **303** is configured to latch a boost signal **316**, which can be used to maintain the global bit line signal RGBL **107** in the GIO circuit **103** based on a previous global bit line signal RGBL **107** from a previous clock cycle (e.g., previous clock period of clock signal CLK **202**). The secondary latch **303** transfers the boost signal **316** to the booster circuit **105**, which is coupled to the global bit line **213**.

(19) The global bit line signal latch **301** includes a first inverter **305** and a second inverter **306**. The input of the first inverter **305** of the global bit line signal latch **301** is coupled to the global bit line **213** and receives the global bit line signal **107**. The first inverter **305** of the global bit line signal latch **301** includes a first enable terminal receiving a latch enable signal LAT **314** and a second enable terminal receiving a latch enable bar signal LATB **315** (e.g., an inverse of the latch enable signal LAT **314**). The latch enable signal LAT **314** and the latch enable bar signal LATB **315** may be generated from a global control block (not shown). The global control block may be located externally from the GIO circuit **103** or within the GIO circuit **103**. The output of the first inverter **305** is coupled to a complimentary global bit line **322**. The second inverter **306** of the global bit line signal latch **301** is coupled to the first inverter **305** of the global bit line signal latch **301**. The second inverter **306** includes a first enable terminal receiving the latch enable bar signal LATB **315** and a second enable terminal receiving the latch enable signal LAT **314**. The input of the second inverter **306** is coupled to the output of the first inverter **305**. The output of the second inverter **306** is coupled to the global bit line **213**.

(20) The global bit line bar signal latch **302** includes a first inverter **307** and a second inverter **308**. The input of the first inverter **307** of the global bit line bar signal latch **302** is coupled to the complimentary global bit line **322** and the output of the first inverter **305** of the global bit line signal latch **301**. The second inverter **308** of the global bit line bar signal latch **302** is coupled to the first inverter **307** of the global bit line bar signal latch **302**. The second inverter **308** of the global bit line bar signal latch **302** includes a first enable terminal receiving the latch enable bar signal LATB **315** and a second enable terminal receiving the latch enable signal LAT **314**. The input of the second inverter **308** receives the output of the first inverter **307**. The output of the second inverter **308** of the global bit line bar signal latch **302** is coupled to the complimentary global bit line **322**, the input of the second inverter **306** of the global bit line signal latch **301**, and the input of the first inverter **307** of the global bit line bar signal latch **302**.

(21) The latch circuit **104** further includes a secondary latch input inverter **309**. The secondary latch input inverter **309** is coupled to the global bit line bar signal latch **302** and receives the output of the first inverter **307** of the global bit line bar signal latch **302**. The output of the secondary latch input inverter **309** is received at a first inverter **310** of the secondary latch **303**. The first inverter **310** of the secondary latch **303** includes a first enable terminal that receives the latch enable bar signal LATB **315**. The first inverter **310** of the secondary latch **303** further includes a second enable terminal that receives the latch enable signal LAT **314**. The output of the first inverter **310** is coupled to an input of a second inverter **311** of the secondary latch **303**. The second inverter **311** receives the output of the first inverter **310** and generates an output that is an inverse of the output of the first inverter **310**. The output of the second inverter **311** of the secondary latch **303** is received by a third inverter **312** of the secondary latch **303**.
(22) The third inverter **312** of the secondary latch **303** includes a first enable terminal receiving the latch enable signal LAT **314** and a second enable terminal receiving the latch enable bar signal LATB **315**. The third inverter **312** of the secondary latch **303** generates a boost signal **316**. The boost signal **316** is stored at a booster node **317** coupled to the output of the third inverter **312** of the secondary latch **303** and the input of the second inverter **311** of the secondary latch **303**. The boost signal **316** is used to drive the global bit line signal RGBL **107** based on a previous global bit line signal RGBL **107** from a previous clock cycle, as described further below.
(23) The booster circuit **105** is coupled to the global bit line signal latch **301**. The GIO circuit **103** may further include a second booster circuit (not shown) that is coupled to the global bit line bar signal latch **302**. The second booster circuit may function similarly to the booster circuit **105** and may drive the global bit line bar signal RGBLB **320** based on a previous global bit line bar signal RGBLB **320** from a previous clock cycle. The booster circuit **105** includes a booster inverter **313**. The input of the booster inverter **313** is coupled to the output of the first inverter **305** of the global bit line signal latch **301**. The input of the booster inverter **313** is also coupled to the input of the second inverter **306** of the global bit line signal latch **301**. The output of the booster inverter **313** is coupled to the output of the second inverter **306** of the global bit line signal latch **301**, the input of the first inverter **305** of the global bit line signal latch **301**, and the global bit line **213**. The booster inverter **313** includes a first enable terminal and a second enable terminal. The first enable terminal and the second enable terminal are coupled to the booster node **317**. The first enable terminal and the second enable terminal each receive the boost signal **316** from the booster node **317**.
(24) FIG. **4** depicts a timing diagram of a read operation of a global bit line, in accordance with some embodiments. The description of the read operation may be best understood when read with reference to FIGS. **2**, **3**, and **4**. At time T=1, bit data containing a logic level of “0” is read from a memory cell **218**. At time t=1, the read global bit line signal RGBL **107** is at logic high (“1”) and the boost signal BOOST **316** is also at logic high (“1”). At the start of a clock cycle, the clock signal CLK **202** may transition from logic low (“0”) to logic high (“1”). The clock signal CLK **202** may alternate between logic low (“0”) and logic high (“1”), for example, based on oscillations of an oscillator (e.g., a quartz crystal) within the memory device **100**. Based on the transition of the clock signal CLK **202** from logic low (“0”) to logic high (“1”), the internal clock signal INT CLK **201** may also transition from logic low (“0”) to logic high (“1”). The internal clock signal INT CLK **201** may be based on the oscillation of, for example, an RC oscillator within the memory device **100**. The internal clock signal INT CLK **201** is generated in the control block of the memory device **100**. Based on the rising edge (e.g., the transition from logic low to logic high) of the internal clock signal INT CLK **201**, the sense amplifier pre-charge signal SAPREB **208** may transition from logic low (“0”) to logic high (“1”). For example, the control block of the memory device **100** transitions the sense amplifier pre-charge signal SAPREB **208** from logic low (“0”) to logic high (“1”) based on the control block detecting the rising edge of the internal clock signal INT CLK **201**. The sense amplifier pre-charge signal SAPREB **208** charges both the read bit line signal RBL **207** and the read bit line bar signal RBLB **206** to logic high (“1”). The internal clock

signal INT CLK **201** may then transition from logic high ("1") to logic low ("0") based on the oscillation of the RC oscillator within the memory device **100**.

(25) Based on the falling edge (e.g., the transition from logic high to logic low) of the internal clock signal INT CLK **201**, the sense amplifier enable signal SAE **209** transitions from logic low to logic high. As discussed above, the falling edge of the internal clock signal INT CLK **201** can be based on the oscillations of an oscillator within the memory device **100**. The control block may sense the falling edge of the internal clock signal INT CLK **201** and switch the sense amplifier enable signal SAE **209** from logic low to logic high. Furthermore, based on the falling edge of the internal clock signal INT CLK **201**, the latch enable signal LAT **314** transitions from logic high to logic low. For example, the control block in the memory device **100** senses the falling edge of the internal clock signal INT CLK **201** and switches the latch enable signal LAT **314** from logic high to logic low. Based on the logic high sense amplifier enable signal SAE **209**, the read bit line signal RBL **207** transitions from logic high ("1") to logic low ("0"). For example, the sense amplifier enable signal SAE **209** drives the read bit line signal RBL **207** to a logic level of "0" based on the logic low bit data read from the memory cell **218**. Furthermore, the read bit line bar signal RBLB **206** remains at logic high ("1"). As discussed above with reference to FIG. **2**, the inverter **210** of the LIO circuit **102** generates a logic high RBLN signal **219** based on the logic low read bit line signal RBL **207**. The logic high RBLN signal **219** is received at the NMOS transistor **212** and couples the global bit line **213** to the VSS voltage level **220**, and the global bit line signal RGBL **107** transitions from logic high to logic low.

(26) As shown in FIG. **3**, the global bit line signal RGBL **107** is received at the first inverter **305** of the global bit line signal latch **301**. The first inverter **305** is enabled by the logic low latch enable signal LAT **314** received at its first enable terminal and the logic high latch enable bar signal LATB **315** received at its second enable terminal. Therefore, the first inverter **305** generates a logic high global bit line bar signal RGBLB **320**, and the global bit line bar signal RGBLB **320** transitions from logic low to logic high. The second inverter **306** of the global bit line signal latch **301** is disabled based on the logic high latch enable bar signal LATB **315** received at its first enable terminal and the logic low latch enable signal LAT **314** received at its second enable terminal.

(27) The first inverter **307** of the global bit line bar signal latch **302** receives the logic high global bit line bar signal RGBLB **320** and generates a logic low global bit line latch signal RGBL LAT **319**. The second inverter **308** of the global bit line bar signal latch **302** is disabled based on the logic high latch enable bar signal LATB **315** received at its first enable terminal and the logic low latch enable signal LAT **314** received at its second enable terminal. Thus, the global bit line bar signal RGBLB **320** is maintained at logic high. The secondary latch input inverter **309** receives the logic low global bit line latch signal RGBL_LAT **319** and generates a logic high secondary latch input signal **318**. The logic high secondary latch input signal **318** is received by the first inverter **310** of the secondary latch **303**. The first inverter **310** of the secondary latch **303** is disabled based on the logic high latch enable bar signal LATB **315** received its first enable terminal and the logic low latch enable signal LAT **314** received at its second enable terminal. Therefore, the boost signal **316** at the booster node **317** is maintained at logic high from the previous clock cycle.

(28) As discussed above, the booster node **317** is coupled to the first enable terminal and the second enable terminal of the booster inverter **313** of the booster circuit **105**. The booster inverter **313** is enabled based on the boost signal **316** received at its first and second enable terminals. Thus the logic high global bit line bar signal RGBLB **320** is received at the input of the booster inverter **313** and the booster inverter **313** outputs a logic low booster inverter output signal **321**. As discussed above, the output of the booster inverter **313** is coupled to the global bit line **213**. The logic low booster inverter output signal **321** and the boost signal **316** can thus drive the global bit line **213** and latch the global bit line signal RGBL **107** based on the previously read data stored in the secondary latch **303** (e.g., the boost signal). When the latch enable signal LAT **314** transitions from logic low to logic high, the first inverter **310** of the secondary latch **303** is enabled based on the

latch enable bar signal LATB **315** received at its first enable terminal and the latch enable signal LAT **314** received at its second enable terminal. The first inverter **310** receives the logic high secondary latch input signal **318** and generates a logic low boost signal **316**, which is latched in the secondary latch **303** based on the second and third inverters (**311**, **312**) of the secondary latch **303**.

(29) At time T=2, as shown in FIG. **4**, bit data containing a logic level of "1" is read from a memory cell **218**. The read global bit line signal **107** is at logic low ("0") and the boost signal **316** is also at logic low ("0"). At the start of a clock cycle, the clock signal CLK **202** may transition from logic low ("0") to logic high ("1"). As discussed above, the clock signal CLK **202** transitions from logic low ("0") to logic high ("1") based on oscillation of an oscillator within the memory device **100**. Based on the transition of the clock signal CLK **202** from logic low ("0") to logic high ("1"), the internal clock signal INT CLK **201** may also transition from logic low ("0") to logic high ("1"). As discussed above, the transition of the internal clock signal INT CLK **201** may be based on the oscillation of an RC oscillator within the memory device **100**. The internal clock signal INT CLK **201** may be generated in the control block of the memory device **100**. Based on the rising edge of the internal clock signal INT CLK **201**, the sense amplifier pre-charge signal SAPREB **208** transitions from logic low ("0") to logic high ("1"). For example, the control block of the memory device **100** may transition the sense amplifier pre-charge signal SAPREB **208** from logic low ("0") to logic high ("1") based on the control block detecting the rising edge of the internal clock signal INT CLK **201**. The sense amplifier pre-charge signal SAPREB charges both the read bit line signal RBL **207** and the read bit line bar signal RBLB **206** to logic high ("1"). The internal clock signal INT CLK **201** may then transition from logic high ("1") to logic low ("0") based on the oscillation of the RC oscillator within the memory device **100**.

(30) Based on the falling edge (e.g., the transition from logic high to logic low) of the internal clock signal INT CLK **201**, the sense amplifier enable signal SAE **209** transitions from logic low to logic high. As discussed above, the control block may sense the falling edge of the internal clock signal INT CLK **201** and switch the sense amplifier enable signal SAE **209** from logic low to logic high. Furthermore, based on the falling edge of the internal clock signal INT CLK **201**, the latch enable signal LAT **314** transitions from logic high to logic low. For example, the control block in the memory device **100** senses the falling edge of the internal clock signal INT CLK **201** and switches the latch enable signal LAT **314** from logic high to logic low. Based on the logic high sense amplifier enable signal SAE **209**, the read bit line signal RBL **207** remains at logic high ("1"). For example, the sense amplifier enable signal SAE **209** can maintain the read bit line signal RBL **207** at the logic high level based on the logic high bit data read from the memory cell **218**. Furthermore, the read bit line bar signal RBLB **206** transitions from logic high ("1") to logic low ("0"). As discussed above with reference to FIG. **2**, the read bit line bar signal RBLB **206** is received at the PMOS transistor **211** of the LIO circuit **102**. The PMOS transistor **211** couples the global bit line **213** to the VDD voltage level **221**, and the global bit line signal RGBL **107** transitions from logic low to logic high.

(31) The first inverter **307** of the global bit line bar signal latch **302** receives the logic low global bit line bar signal RGBLB **320** and generates a logic high global bit line latch signal RGBL_LAT **319**. The second inverter **308** of the global bit line bar signal latch **302** is disabled based on the logic high latch enable bar signal LATB **315** received at its first enable terminal and the logic low latch enable signal LAT **314** received at its second enable terminal. Thus, the global bit line bar signal RGBLB **320** is maintained at logic high. The secondary latch input inverter **309** receives the logic high global bit line latch signal RGBL_LAT **319** and generates a logic low secondary latch input signal **318**. The logic low secondary latch input signal **318** is received by the first inverter **310** of the secondary latch **303**. The first inverter **310** of the secondary latch **303** is disabled based on the logic high latch enable bar signal LATB **315** received its first enable terminal and the logic low latch enable signal LAT **314** received at its second enable terminal. Therefore, the boost signal **316** at the booster node **317** is maintained at logic low from the previous clock cycle.

(32) The first inverter **307** of the global bit line bar signal latch **302** receives the logic low global bit line bar signal RGBLB **320** and generates a logic high global bit line latch signal RGBL_LAT **319**. The second inverter **308** of the global bit line bar signal latch **302** is disabled based on the logic high latch enable bar signal LATB **315** received at its first enable terminal and the logic low latch enable signal LAT **314** received at its second enable terminal. Thus, the global bit line bar signal RGBLB **320** is maintained at logic high. The secondary latch input inverter **309** receives the logic high global bit line latch signal RGBL_LAT **319** and generates a logic low secondary latch input signal **318**. The logic low secondary latch input signal **318** is received by the first inverter **310** of the secondary latch **303**. The first inverter **310** of the secondary latch **303** is disabled based on the logic high latch enable bar signal LATB **315** received its first enable terminal and the logic low latch enable signal LAT **314** received at its second enable terminal. Therefore, the boost signal **316** at the booster node **317** is maintained at logic low from the previous clock cycle.
(33) As discussed above, the booster node **317** is coupled to the first enable terminal and the second enable terminal of the booster inverter **313** of the booster circuit **105**. The booster inverter **313** is enabled based on the boost signal **316** received at its first and second enable terminals. Thus, the logic low global bit line bar signal RGBLB **320** is received at the input of the booster inverter **313** and the booster inverter **313** outputs a logic high booster inverter output signal **321**. As discussed above, the output of the booster inverter **313** is coupled to the global bit line **213**. The logic high booster inverter output signal **321** and the boost signal **316** can thus drive the global bit line **213** and latch the logic high global bit line signal RGBL **107** based on the previously read data stored in the secondary latch **303** (e.g., the boost signal **316**). When the latch enable signal LAT **314** transitions from logic low to logic high, the first inverter **310** of the secondary latch **303** is enabled based on the latch enable bar signal LATB **315** received at its first enable terminal and the latch enable signal LAT **314** received at its second enable terminal. The first inverter **310** receives the logic low secondary latch input signal **318** and generates a logic high boost signal **316**, which is latched in the secondary latch **303** based on the second and third inverters (**311**, **312**) of the secondary latch **303**.
(34) The systems and methods described above can improve the slope of the logic levels of the global bit line signal RGBL **107** by driving the global bit line signal RGBL **107** with components within the GIO circuit **103**. Furthermore, the disclosed systems and methods can eliminate or mitigate the dependency of latching the global bit line signal RGBL **107** on the sense amplifier enable signal SAE **209** being enabled. The global bit line signal RGBL **107** may be latched in the GIO circuit **103** based on signals stored in the GIO circuit **103**. Therefore, the pulse width (e.g., the time at logic high) of the sense amplifier enable signal SAE **209** may be decreased, and the pre-charging of the read bit line signal RBL **207** and the read bit line bar signal RBLB **206** may occur sooner than if latching the global bit line signal RGBL **107** depended entirely on the sense amplifier enable signal SAE **209** driving the global bit line signal RGBL **107** to a sufficient logic level. This can lead to decreased access time of bit data from the memory cell array **101** and decreased read cycle times.
(35) FIG. **5** depicts a method, in accordance with some embodiments. In the example embodiment depicted in FIG. **5**, the method **500** includes a first step **501** of receiving a global bit line signal on a global bit line during a current clock cycle. The first step **501** is depicted in FIG. **2**. The GIO circuit **103** receives the global bit line signal **107** on the global bit line **213**. A second step **502** in the method **500** is generating a boost signal based on a previous global bit line signal generated during a previous clock cycle. The second step **502** is shown in FIG. **3**. The boost signal **316** is generated in the secondary latch **303** based on a previous global bit line signal generated during a previous clock cycle, as described further above with respect to FIG. **3**. A third step **503** in the method **500** is driving the global bit line signal based on the boost signal. The third step **503** is shown in FIG. **3**. The boost signal **316** is received by the booster inverter **313**. The booster inverter **313** drives the global bit line signal **107** on the global bit line **213**.

(36) Systems and methods are disclosed herein. In one example, a memory device includes a first memory array, a local input/output (LIO) circuit, and a global input/output (GIO) circuit. The first memory array includes a memory cell and a local bit line. The LIO circuit is coupled to the local bit line and is configured to receive a local bit line signal on the local bit line and to generate a global bit line signal on a global bit line based on the local bit line signal. The GIO circuit is coupled to the LIO circuit and is configured to receive the global bit line signal. The GIO circuit comprises a latch circuit including a global bit line signal latch that is configured to latch the global bit line signal. The GIO circuit further includes a booster circuit coupled to the global bit line signal latch that is configured to drive the global bit line signal in the GIO circuit based on a previous global bit line signal generated at a previous clock cycle.
(37) In another example, a global input/output (GIO) circuit is configured to receive a global bit line signal on a global bit line. The GIO circuit comprises a global bit line signal latch coupled to the global bit line. The global bit line signal latch is configured to latch the global bit line signal. The GIO circuit further comprises a global bit line bar signal latch is coupled to the global bit line signal latch. The global bit line bar signal latch is configured to latch a global bit line bar signal. The GIO circuit further comprises a booster circuit coupled to the global bit line. The booster circuit is configured to receive a boost signal and to drive the global bit line signal in the GIO circuit based on the boost signal.
(38) In another example, a method includes receiving a global bit line signal on a global bit line during a current clock cycle. The method further includes generating a boost signal based on a previous global bit line signal generated during a previous clock cycle. The method further includes driving the global bit line signal based on the boost signal.
(39) It will be appreciated by those skilled in the art that changes could be made to the embodiments described above without departing from the broad inventive concept thereof. It is understood, therefore, that the invention disclosed herein is not limited to the particular embodiments disclosed, and is intended to cover modifications within the spirit and scope of the present invention.

Claims

1. A memory device comprising: a first memory array including a memory cell and a local bit line; a local input/output (LIO) circuit coupled to the local bit line, the LIO circuit configured to receive a local bit line signal on the local bit line and to generate a global bit line signal on a global bit line based on the local bit line signal; and a global input/output (GIO) circuit coupled to the LIO circuit, the GIO circuit configured to receive the global bit line signal, the GIO circuit comprising: a latch circuit including a global bit line signal latch, the global bit line signal latch configured to receive and latch the global bit line signal; and a booster circuit coupled to the global bit line signal latch, an output of the booster circuit configured to drive the global bit line signal in the GIO circuit based on a previous global bit line signal generated at a previous clock cycle.
2. The memory device of claim 1, the latch circuit further comprising a secondary latch including a booster node, the booster node configured to latch a boost signal based on the previous global bit line signal.
3. The memory device of claim 2, wherein the booster circuit is further coupled to the secondary latch, wherein the booster circuit comprises a booster inverter configured to receive a global bit line bar signal and the boost signal, the global bit line bar signal being an inverse of the global bit line signal.
4. The memory device of claim 3, wherein the booster inverter is enabled based on the boost signal.
5. The memory device of claim 1, wherein the latch circuit further comprises a global bit line bar signal latch configured to latch a global bit line bar signal, the global bit line bar signal being an inverse of the global bit line signal.

6. The memory device of claim 1, wherein the global bit line signal latch comprises a plurality of inverters, wherein one or more of the plurality of inverters receives a latch enable signal and is enabled or disabled based on the latch enable signal.
7. The memory device of claim 6, further comprising a global control block coupled to the GIO circuit, the global control block configured to generate the latch enable signal.
8. The memory device of claim 1, wherein the memory device is a static random-access memory (SRAM) device.
9. The memory device of claim 1, wherein the LIO circuit is configured to generate a read bit line signal, wherein the global bit line signal is generated based on the read bit line signal.
10. The memory device of claim 9, wherein the LIO circuit further comprises a sense amplifier configured to charge the read bit line signal to a sufficient logic level.
11. A global input/output (GIO) circuit configured to receive a global bit line signal on a global bit line, the GIO circuit comprising: a global bit line signal latch coupled to the global bit line, the global bit line signal latch configured to latch the global bit line signal; a global bit line bar signal latch coupled to the global bit line signal latch, the global bit line bar signal latch configured to latch a global bit line bar signal, the global bit line bar signal being an inverse of the global bit line signal; and a booster circuit coupled to the global bit line, the booster circuit configured to receive a boost signal and the global bit line bar signal and drive the global bit line signal in the GIO circuit based on the boost signal.
12. The GIO circuit of claim 11, further comprising a secondary latch coupled to the global bit line bar signal latch, the secondary latch configured to generate the boost signal based on a previous global bit line signal from a previous clock cycle.
13. The GIO circuit of claim 11, wherein the booster circuit comprises a boosting inverter, the boosting inverter being enabled or disabled based on the boost signal.
14. The GIO circuit of claim 11, wherein the global bit line signal latch comprises a plurality of inverters, wherein one or more of the plurality of inverters is configured to receive a latch enable signal and is enabled or disabled based on the latch enable signal.
15. A method comprising: receiving a global bit line signal on a global bit line during a current clock cycle; generating a global bit line bar signal being an inverse of the global bit line signal; generating a boost signal based on a previous global bit line signal generated during a previous clock cycle; and driving the global bit line signal based on the boost signal and the global bit line bar signal.
16. The method of claim 15 further comprising receiving the boost signal at a booster circuit including a booster inverter, the booster inverter configured to drive the global bit line signal.
17. The method of claim 15, further comprising latching the global bit line signal in a global bit line signal latch.
18. The method of claim 15, further comprising latching the global bit line bar signal in a global bit line bar signal latch.
19. The method of claim 15, wherein the boost signal is generated at a secondary latch comprising a plurality of inverters.
20. The method of claim 19, wherein one or more of the plurality of inverters is a first inverter, the first inverter being enabled or disabled based on receiving a latch enable signal.